Assessing Time-Competitive Cohesion in the Rail O/D Network
: A Modified k-core Decomposition Approach

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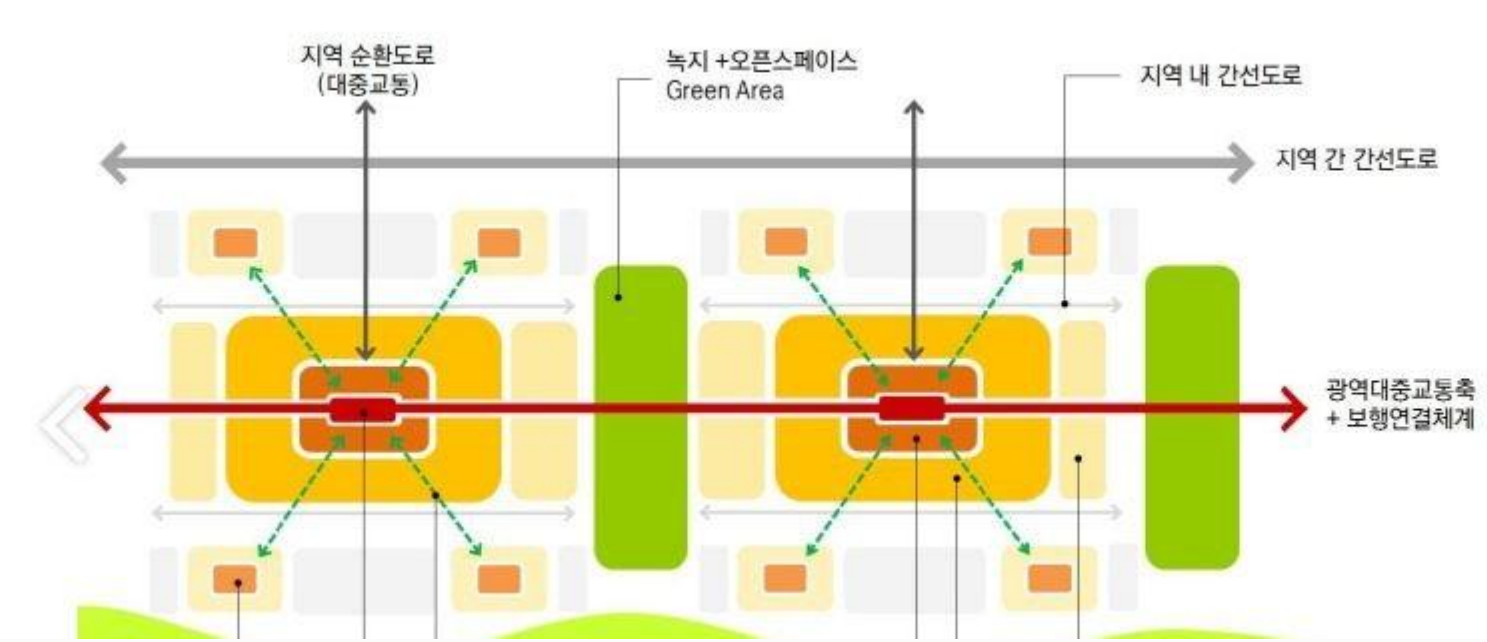
High-quality trunk public transit is necessary for TOD

- Many studies describe **high-quality trunk public transit is necessary for TOD**.
  - *Parcels surrounding transit stations gain value because* ***they enjoy better connectivity to the region****—i.e., residents can more easily and conveniently reach jobs, shops, and other destinations; more potential shoppers pass by retail outlets… (World Bank)*
  - *Walkable access to* ***rapid and frequent transit****, defined as rail transit or bus rapid transit (BRT), is integral to the TOD concept and a prerequisite for TOD Standard recognition (The TOD Standard v2.1, 2017)*

High-quality trunk public transit is necessary for TOD

- A locality can only be restructured around transit if transit is sufficiently competitive relative to private cars.
- Therefore, it is crucial to assess how competitive public transit is compared with car travel.
- For the design of TOD, micro-scale urban design is important, but network analysis from a broader, regional perspective is also essential.



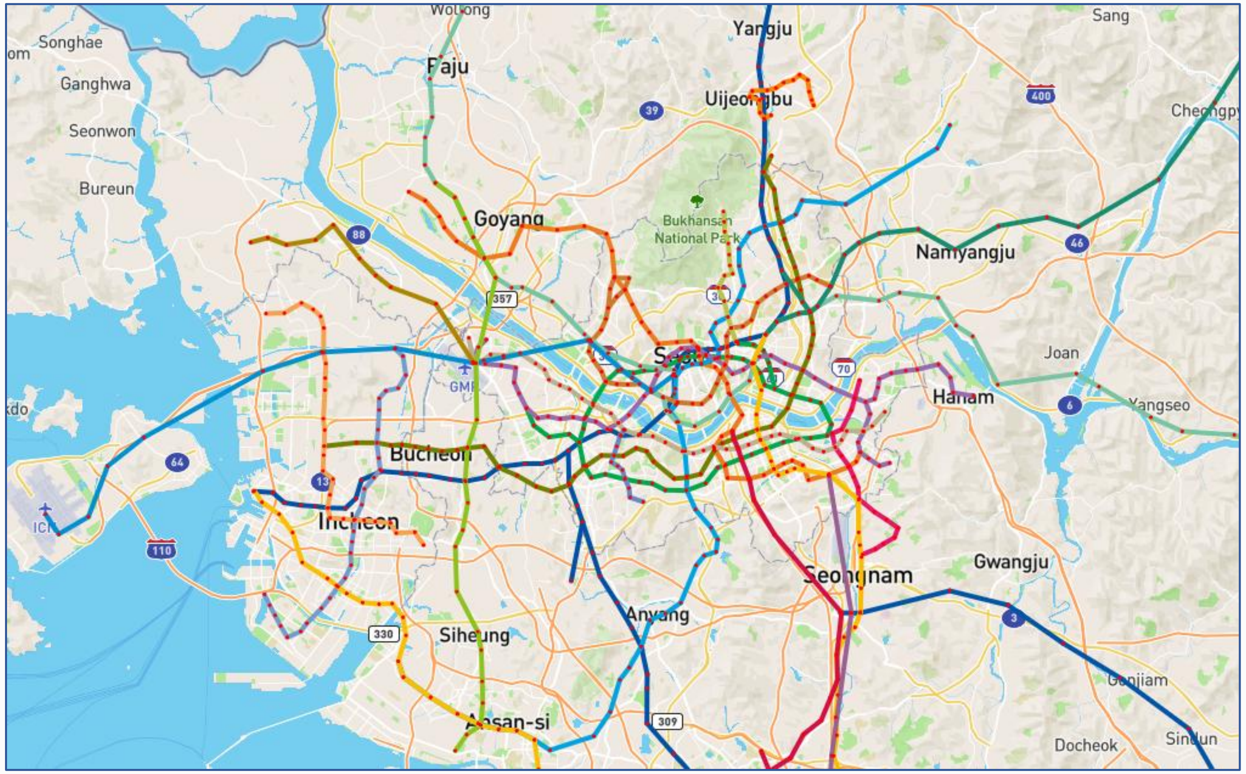
High-quality trunk public transit is necessary for TOD

- The more destinations that can be reached competitively by public transit rather than by private car, **the more competitive transit becomes and the more opportunities** there are to use it.
- From an O/D network perspective, transit should have **a wide and stable range of destinations and trip markets** in which it is time-competitive.
- We define a **Time-Competitive Rail Opportunity Network (TCRON)** and combine biconnectivity with activation-shell decomposition to ask:
  - ✓ To what extent is the metropolitan region coherently bound together by competitive rail connections?
  - ✓ Which areas are stably included in this structure, and where is connectivity fragile or precarious?

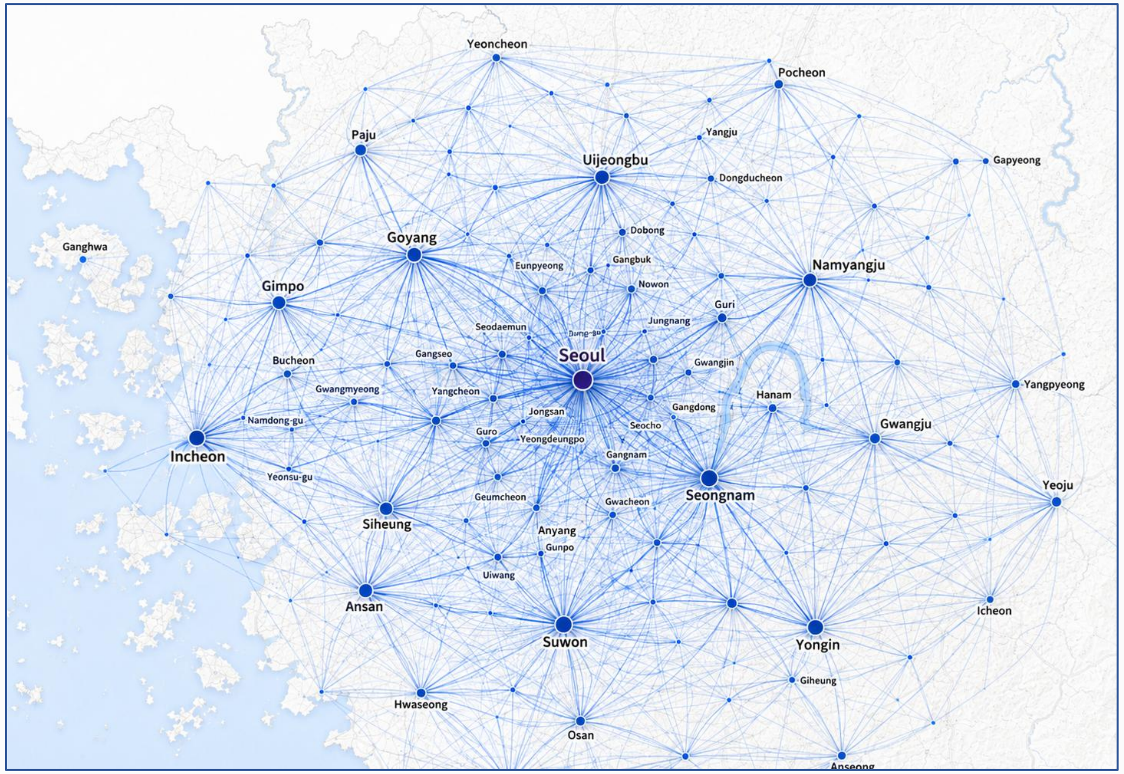
The concept of time-compatitive rail opportunity network (TCRON)

- TCRON is a complex network (O/D network) constructed from rail′s time competitiveness relative to car.
- Nodes represent specific regions (not stations) in the Seoul metropolitan area.
- Links exist only where rail is competitive for travel between nodes.

Real rail network

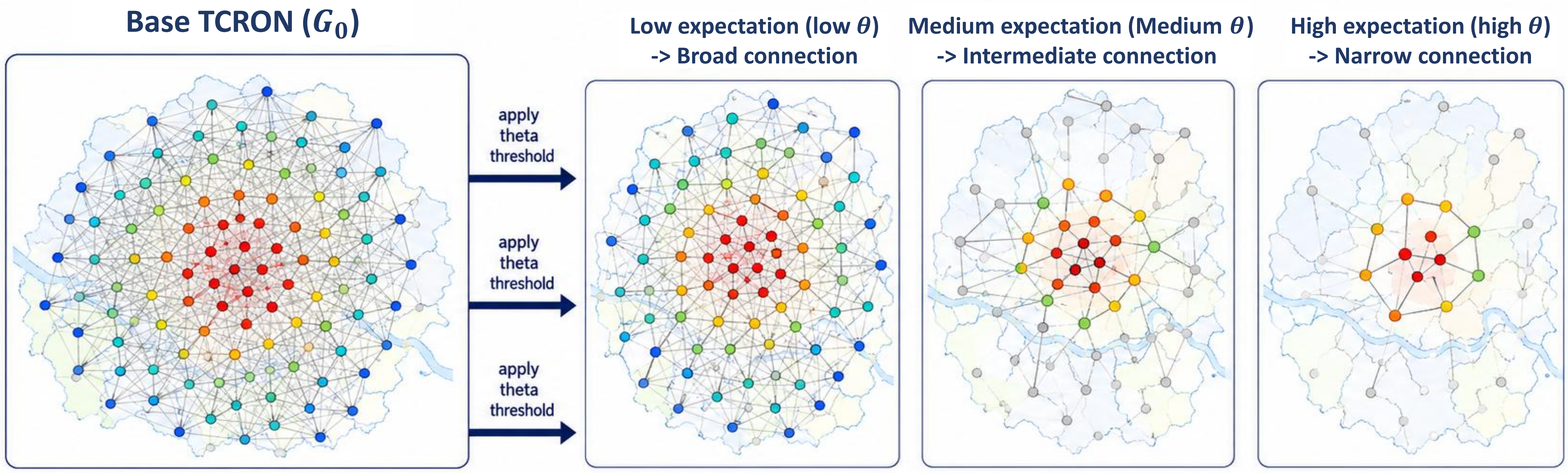


TCRON



TCRON contraction under stricter time-competitiveness expectations

- Rail users perceive the TCRON as different sub-networks depending on their time-competitiveness expectation (θ).
- The higher a user's time-competitiveness expectation (θ), the smaller the TCRON becomes for that user.



Mathematical definition of TCRON

- TCRON retains only OD pairs whose rail competitiveness exceeds a threshold, meaning that rail opportunities remain practically available.

$$e = \{i, j\} \qquad C_e = \frac{T_e^{road}}{T_e^{rail}}$$

$$E_\theta = \{e \in E : C_e \geq \theta\} \qquad G_\theta = (V, E_\theta)$$

$$\theta_c : threshold\ value\ for\ railway\ travel\ time\ competitiveness$$

Giant cluster size and functional efficiency of the TCRON

- By measuring either the number of OD pairs that remain in the TCRON or the associated rail demand (D_{ij}), we can assess how competitively rail binds the metropolitan region together.

- A large giant cluster size (GCS) indicates that rail is competitive across a wide range of trip markets and that a substantial volume of travel demand is included.

- A high functional efficiency (FE) value implies that the TCRON forms a functionally cohesive competitive opportunity field.

$$D_i(\theta) = \sum_{j \neq i} D_{ij} \cdot \mathbb{1}\left(C_{ij} \geq \theta\right)$$

$$GCS(\theta) = \sum_{i \in \mathrm{L}_\theta} D_i(\theta)$$

$$FE(\theta) = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{1}{T_{ij}^{rail}(\theta)}$$

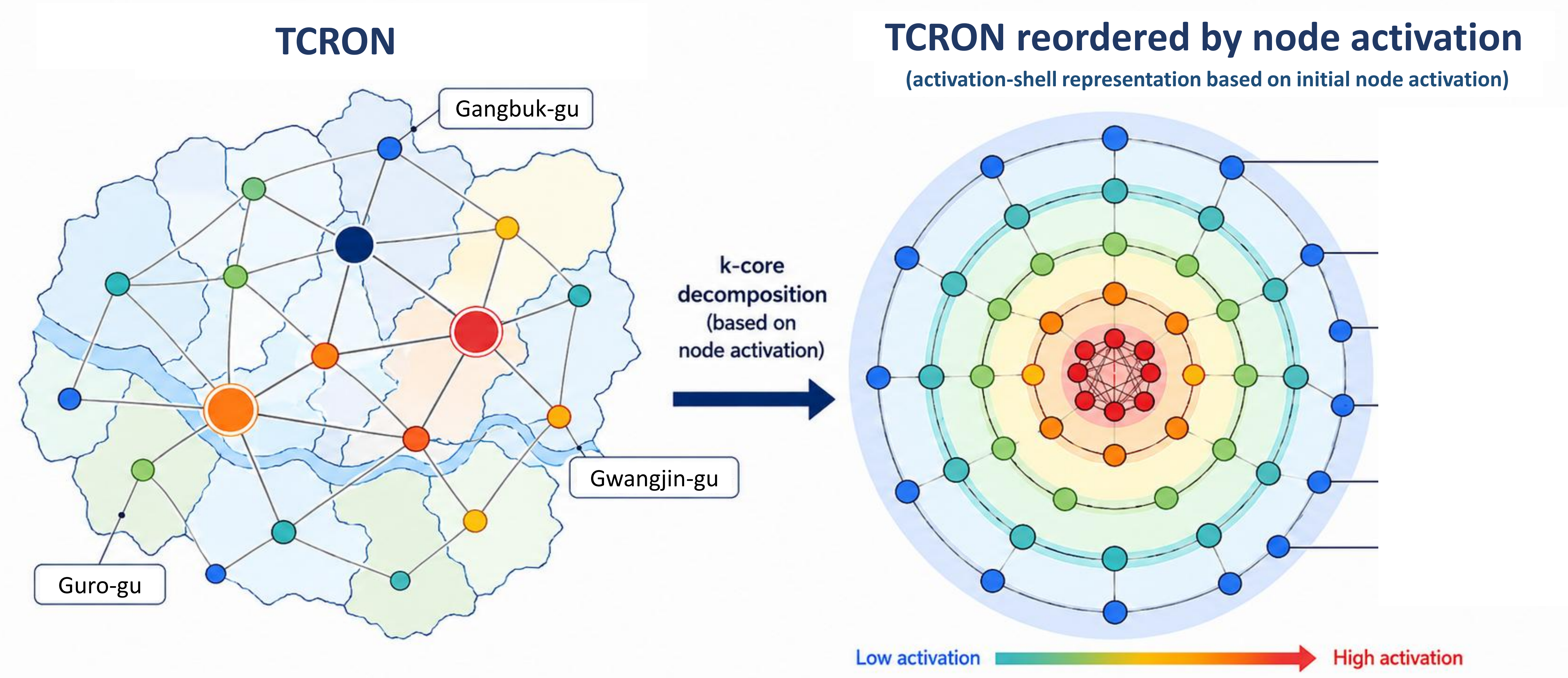
Modified k-core decomposition

- K-core decomposition is a process of iteratively peeling off graph layers by removing nodes with the lowest degree (Alvarez-Hamelin et al., 2005).
- In this study, topological degree is replaced by node activation, and the algorithm proceeds as follows.

$$a_i(\theta) = \frac{D_i(\theta)}{D_i(0)} \qquad D^G_{\boldsymbol{S_k}} = \frac{\Delta S^G_{S_k}}{n_{\boldsymbol{S_k}}}$$

1. $C_k = \{v \in V \mid \mathrm{a_G(v)} \geq k\}$: the k-core subgraph induced by nodes with node activation at least k.
2. $\kappa(v) = max\{k \mid v \in C_k\}$: coreness of node v, i.e., the highest k-core it belongs to.
3. K-shell: $S_k = C_k \backslash C_{k+1}$, the nodes in the k-core but not (k+1)-core, iteratively removed in shells from the periphery inward.

Reordering the TCRON by node activation



K-core decomposition: Peeling the network like an onion

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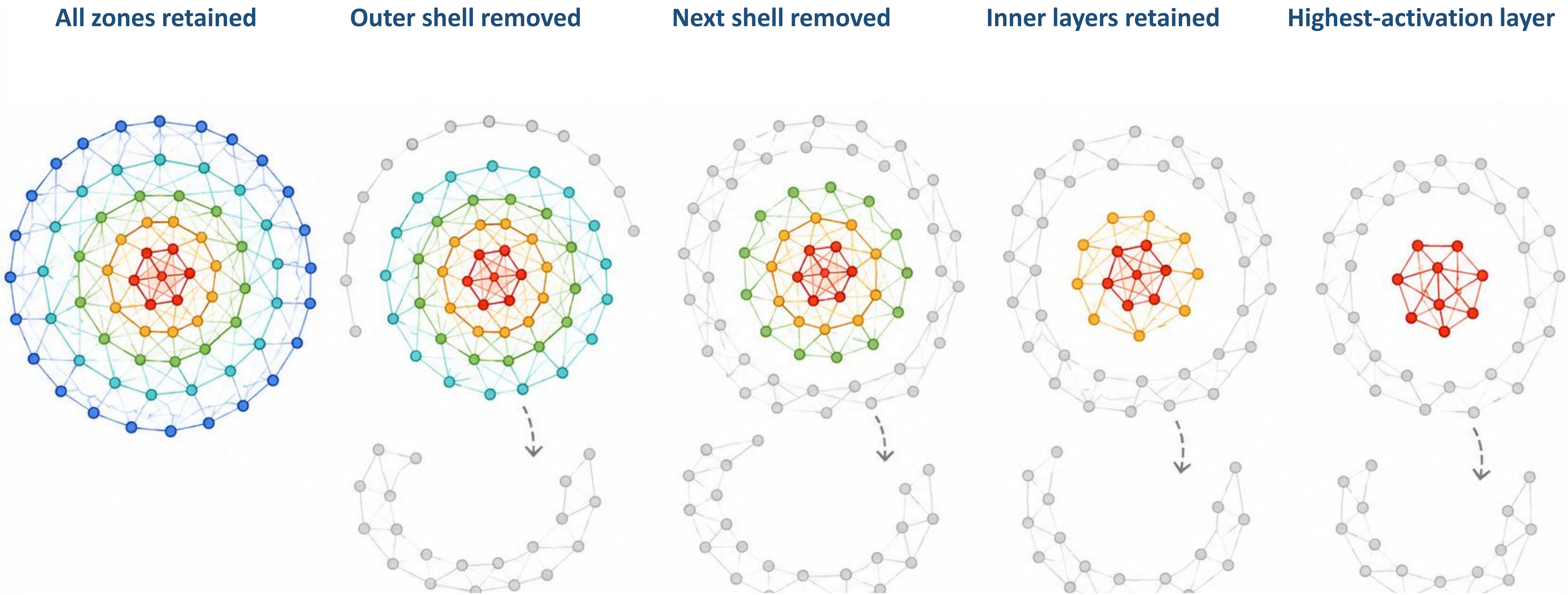
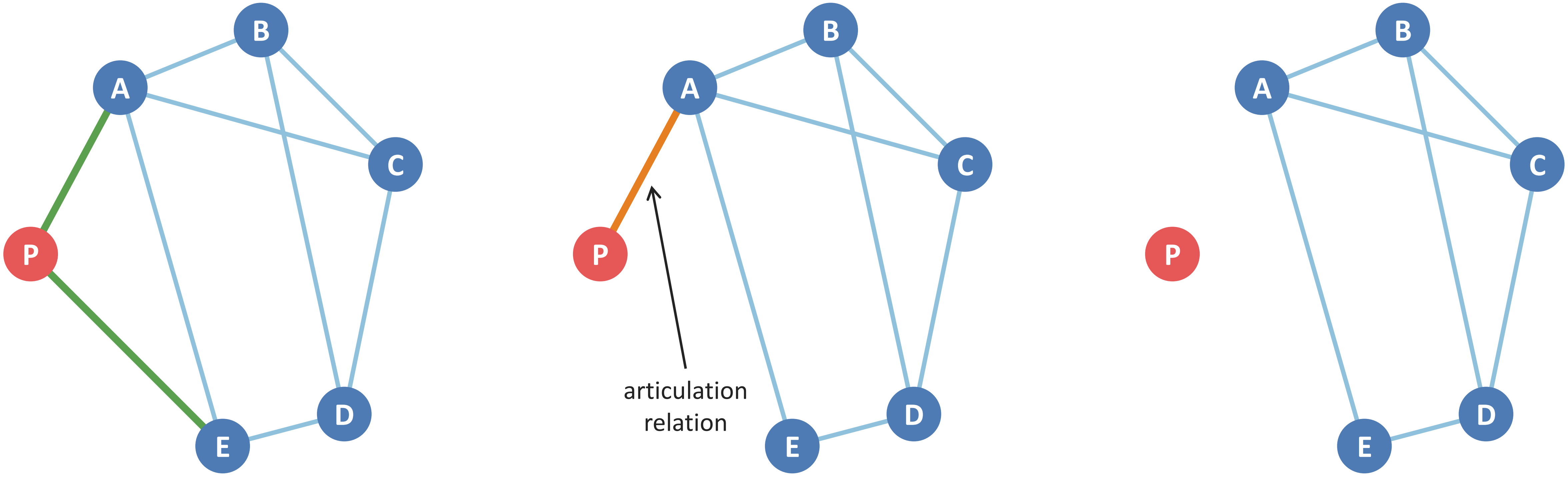


Lower activation (outer layers) → Higher activation (inner layers)

Gray nodes: removed at the current alpha cutoff | Colored nodes: retained

Connected inclusion is not redundant embeddedness

- COC: a connected component of $G(\theta)$
- LCOC: the COC with the greatest internal rail-demand mass
- LBOC: the largest biconnected block inside the LCOC
- Tethered inclusion: in the LCOC but outside the LBOC through an articulation structure



Redundantly embedded

Contingently included, tethered (single-link dependence)

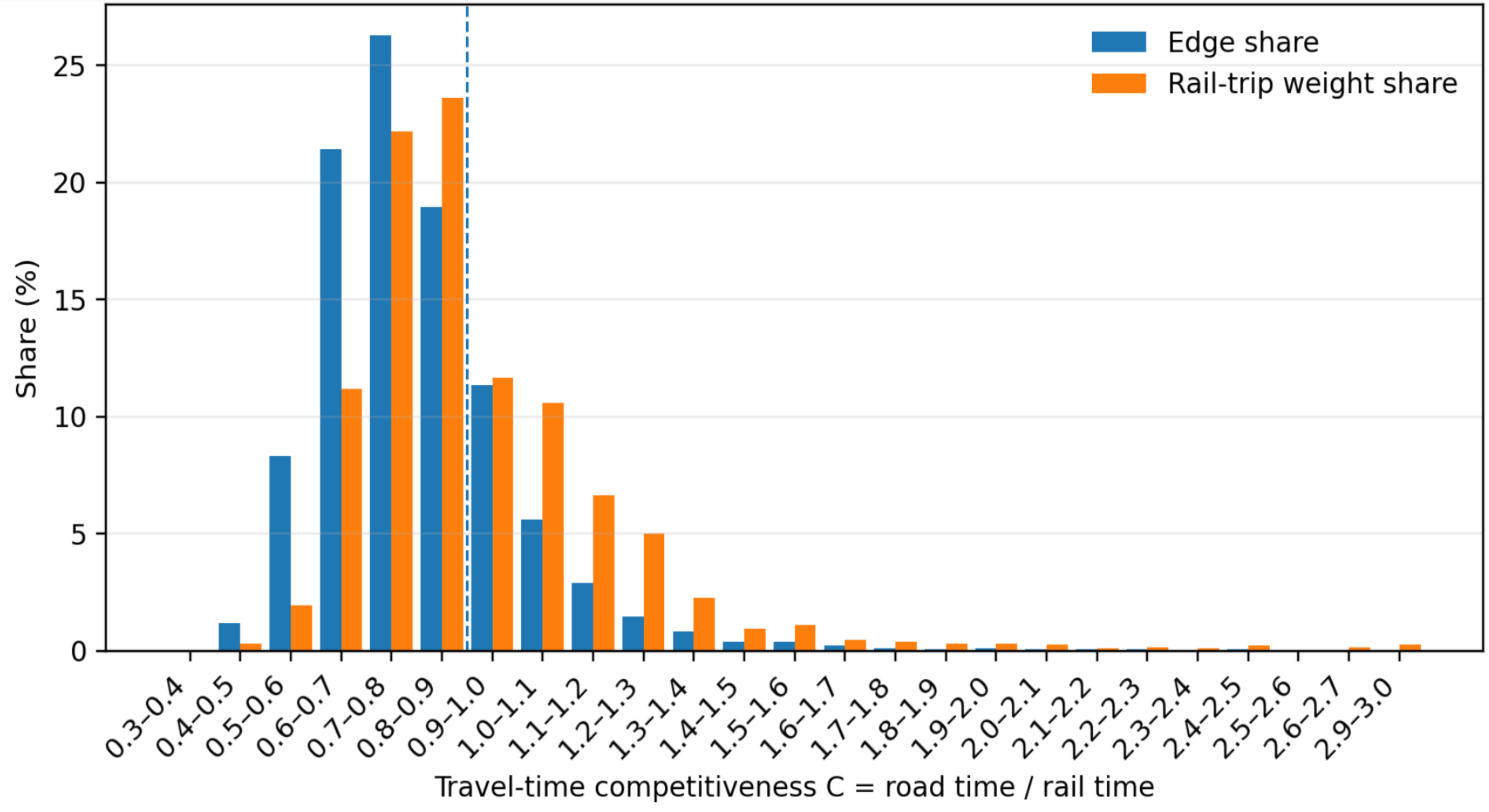
Separated

Data and study area

- The case study focuses on the Seoul Metropolitan Area in South Korea in 2025.
- The analysis combines nationally constructed road and rail networks, OD data, and individual travel survey data.

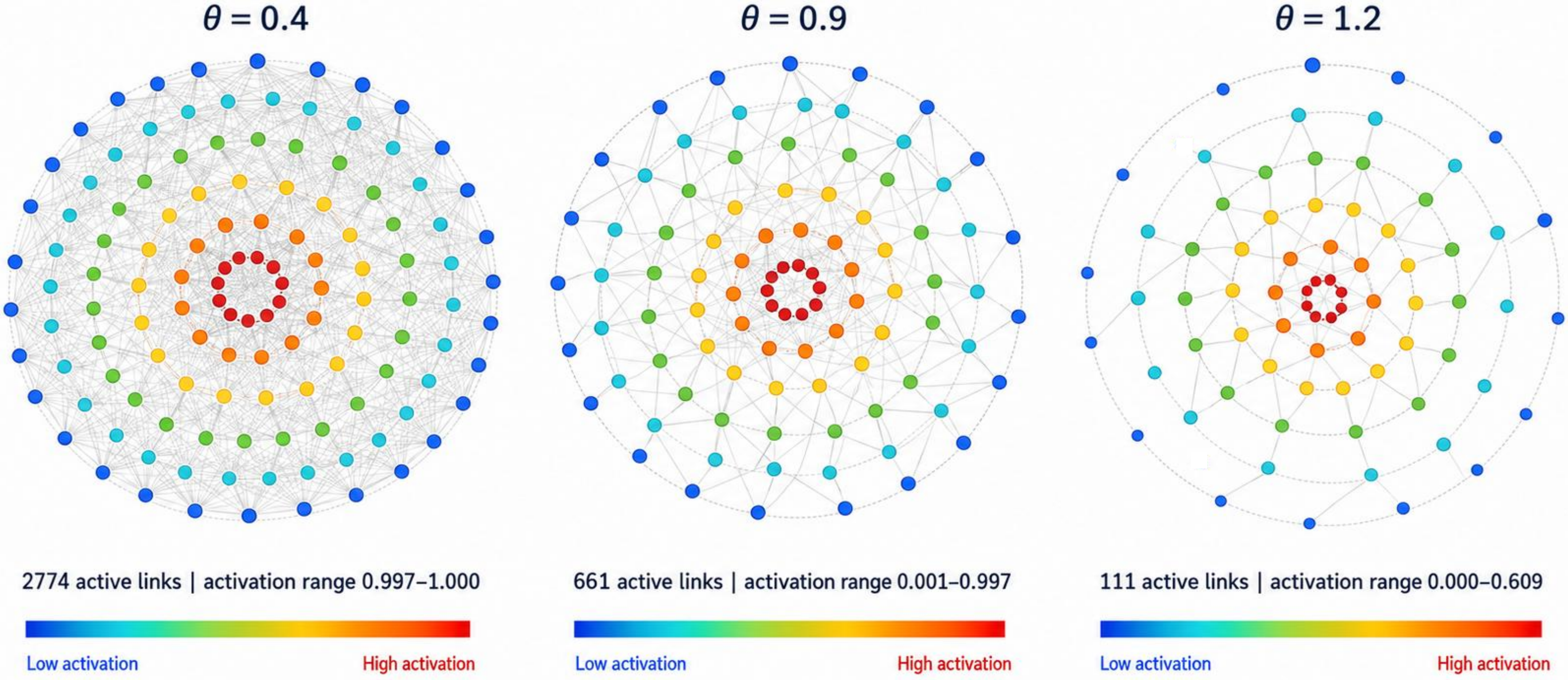
Data Input	Explanation	Use
Rail and road networks	Seoul metropolitan area	Calculating rail / road travel time
Rail and road OD volumes	Mode-specific demand, 75 zones; 2,775 undirected OD pairs	Observed weights and rail share
Rail / road travel time	Computed from the network and O/D data	Assessment of rail competitiveness
Auxiliary time, distance, number of boardings		Access, detour, and transfer impedances

Competitiveness distribution of G_0

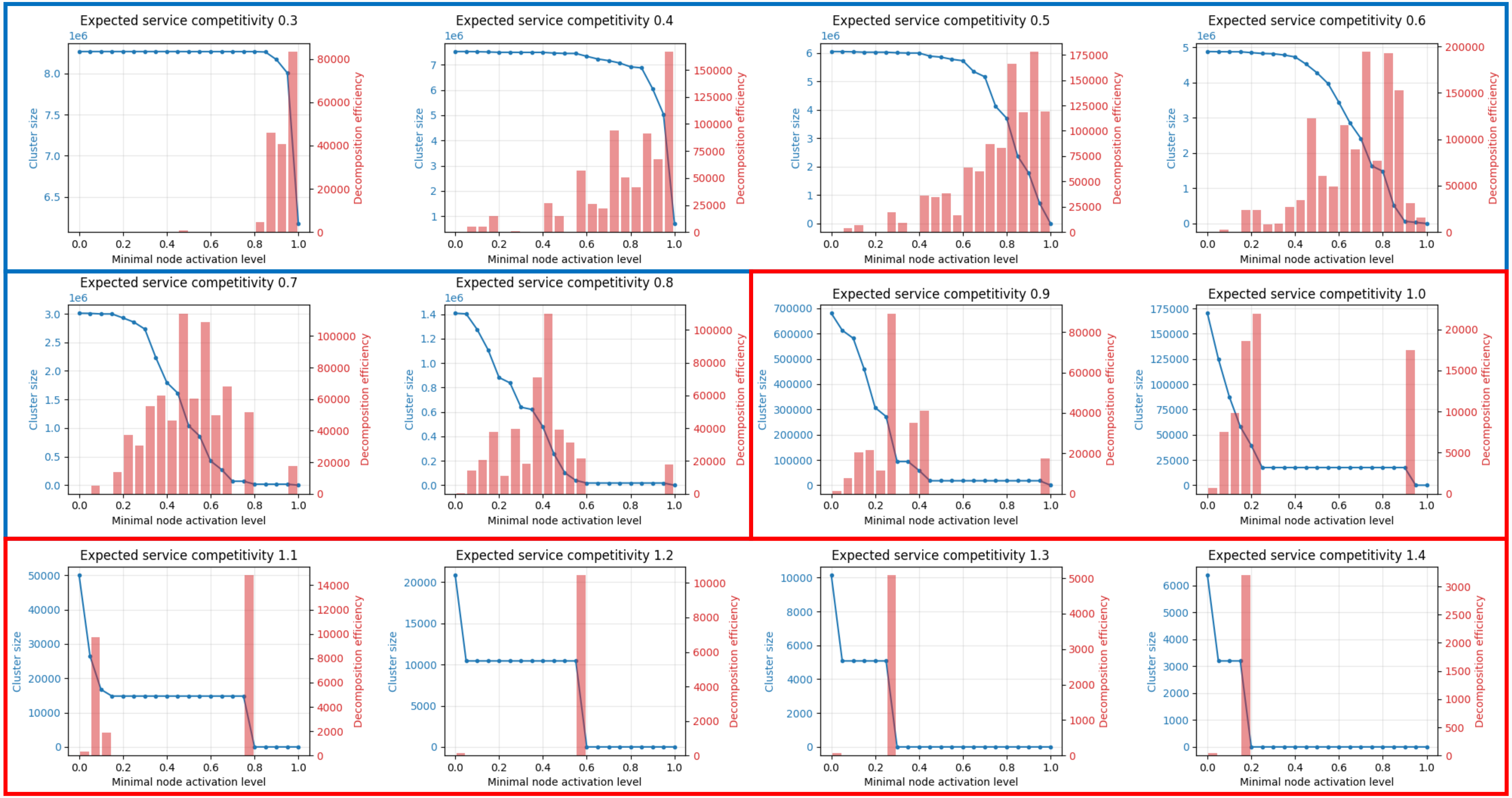


Statistics		**Values**
Number of edges		2,775
C	Average	0.808
	Median	0.770
	Maximum	2.998

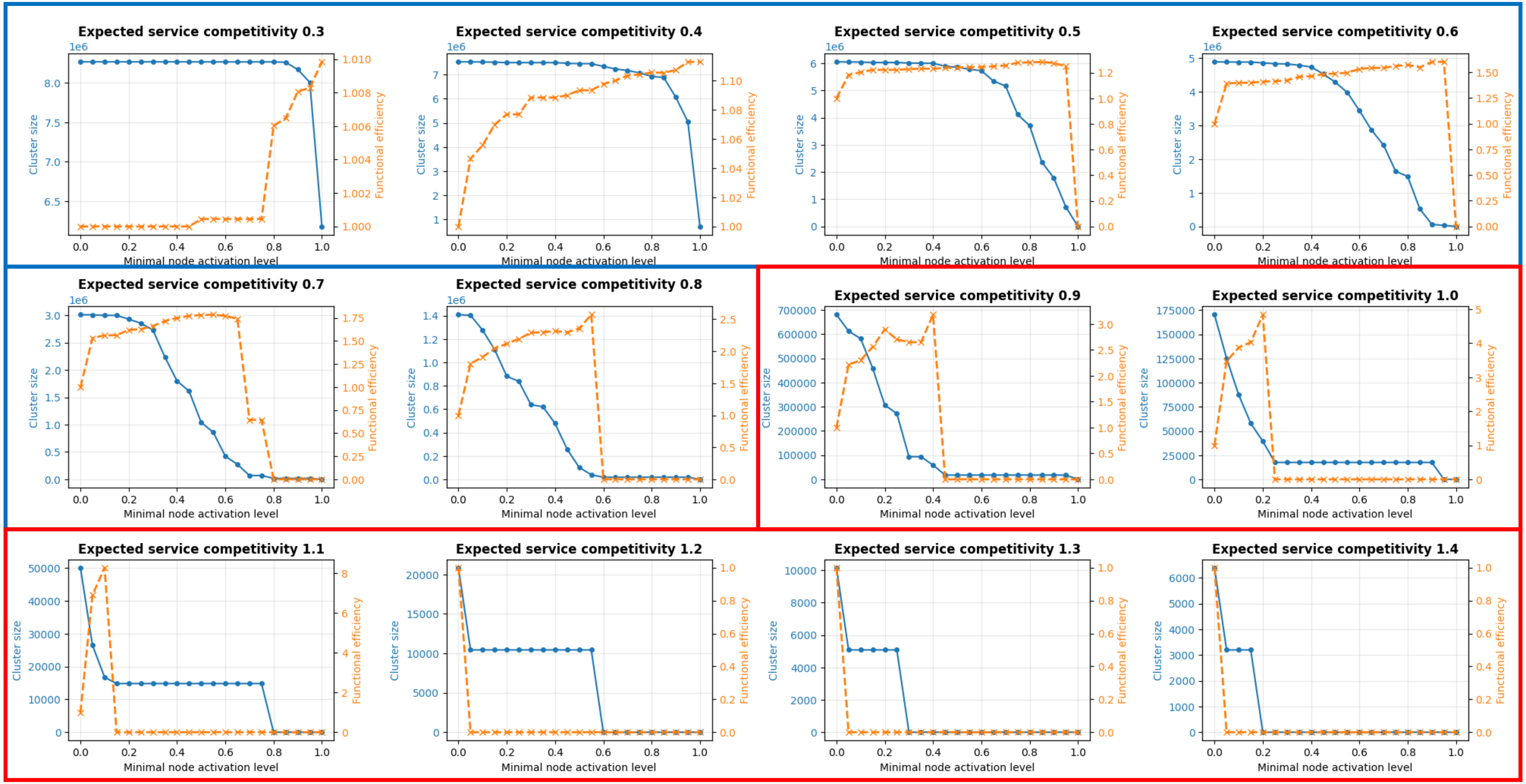
Onion-structure of TCRON reordered by node activation



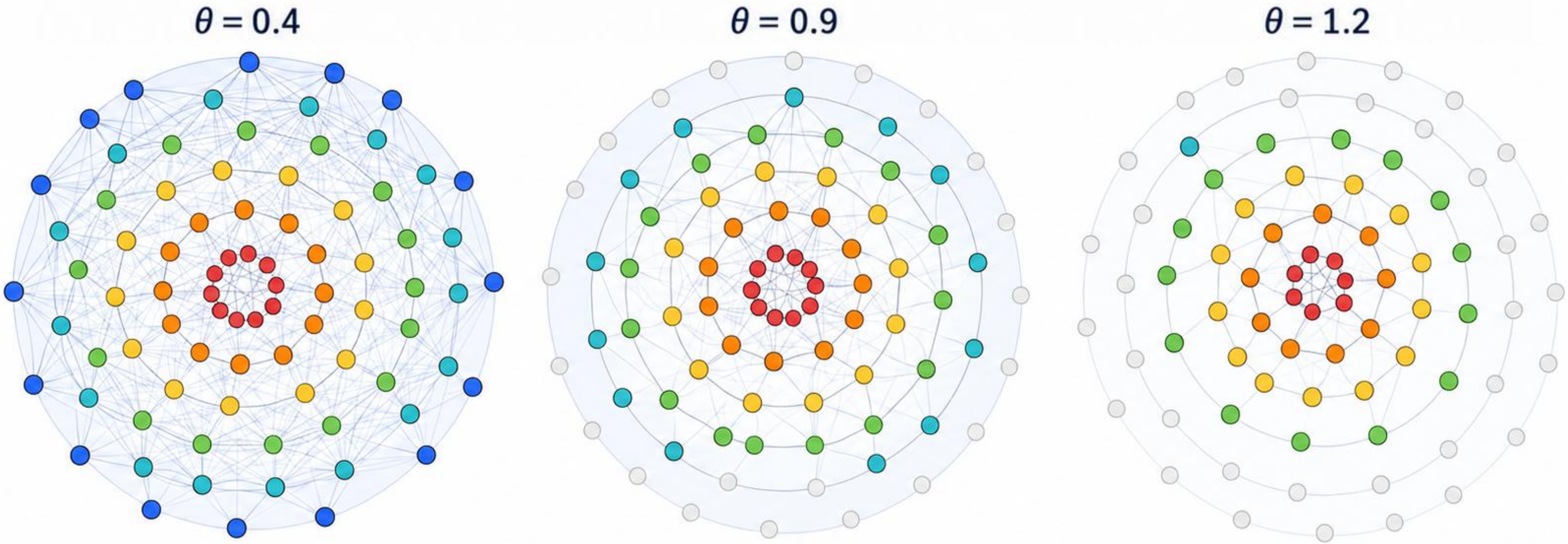
Robustness test under increasing θ_c



Functional efficiency test under increasing θ_c



Pre-collapse core identified by the k-core decompostiion



keep 73 zones and 2628 links;
largest connected core: 73 zones

keep 55 zones and 502 links;
largest connected core: 55 zones

keep 51 zones and 91 links;
largest connected core: 47 zones

Thresholding result: sparse but demand-concentrated

θ	Edges	Edge retention	Rail-trip retention	Purpose-demand retention	LCOC zones	LBOC zones	FE
0.50	2,741	98.8%	99.7%	99.4%	75	75	0.994
0.70	1,916	69.0%	86.6%	85.1%	75	75	0.845
0.85	886	31.9%	52.4%	60.5%	75	75	0.658
0.90	661	23.8%	40.8%	52.0%	75	74	0.607
0.95	480	17.3%	34.0%	46.4%	75	73	0.547
1.10	191	6.9%	18.6%	28.7%	69	59	0.356
1.30	71	2.6%	6.9%	15.4%	44	11	0.096

23.8%
relations remain at θ=0.9

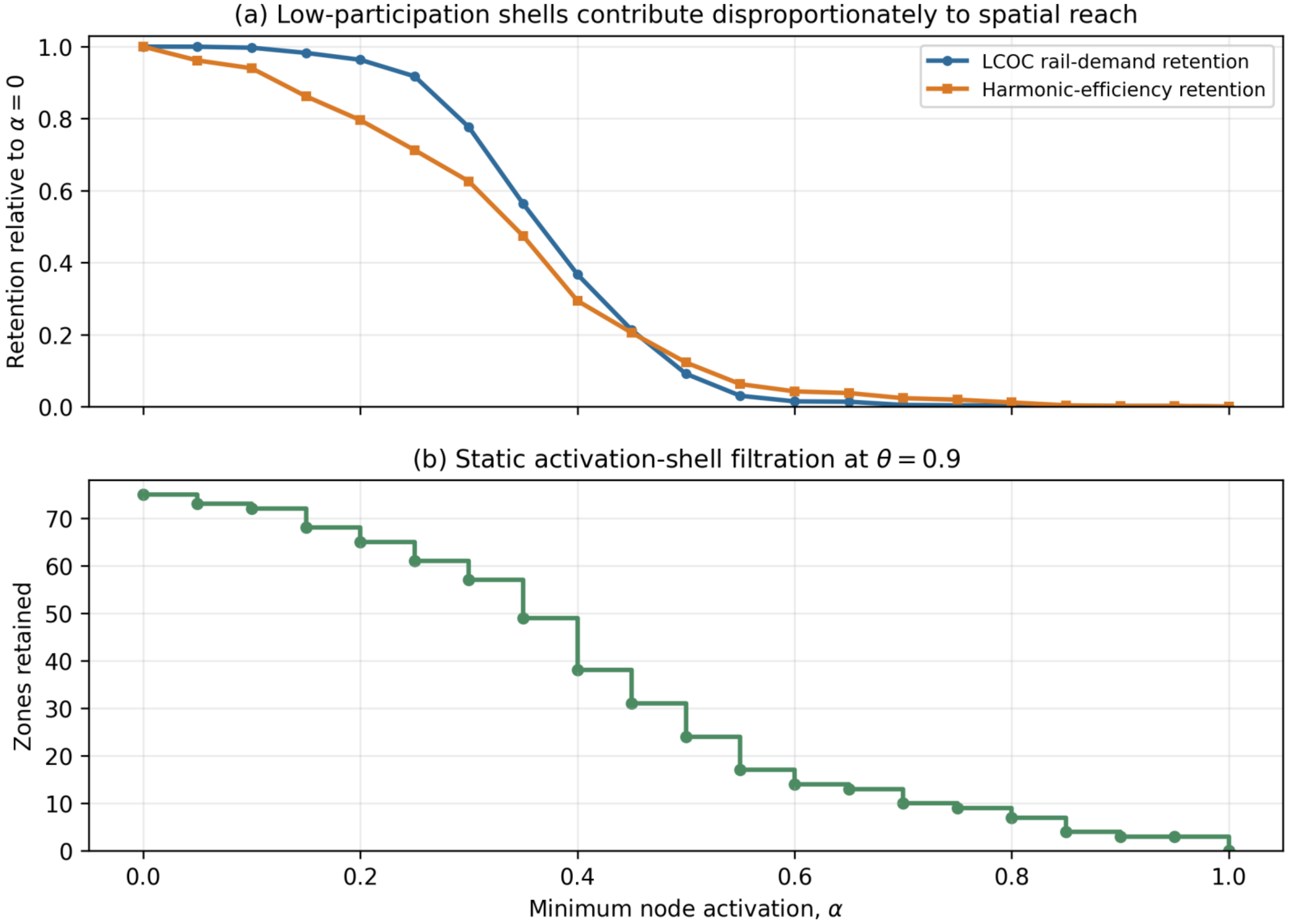
40.8%
rail trips retained

52.0%
purpose demand retained

75 / 74
LCOC / LBOC zones

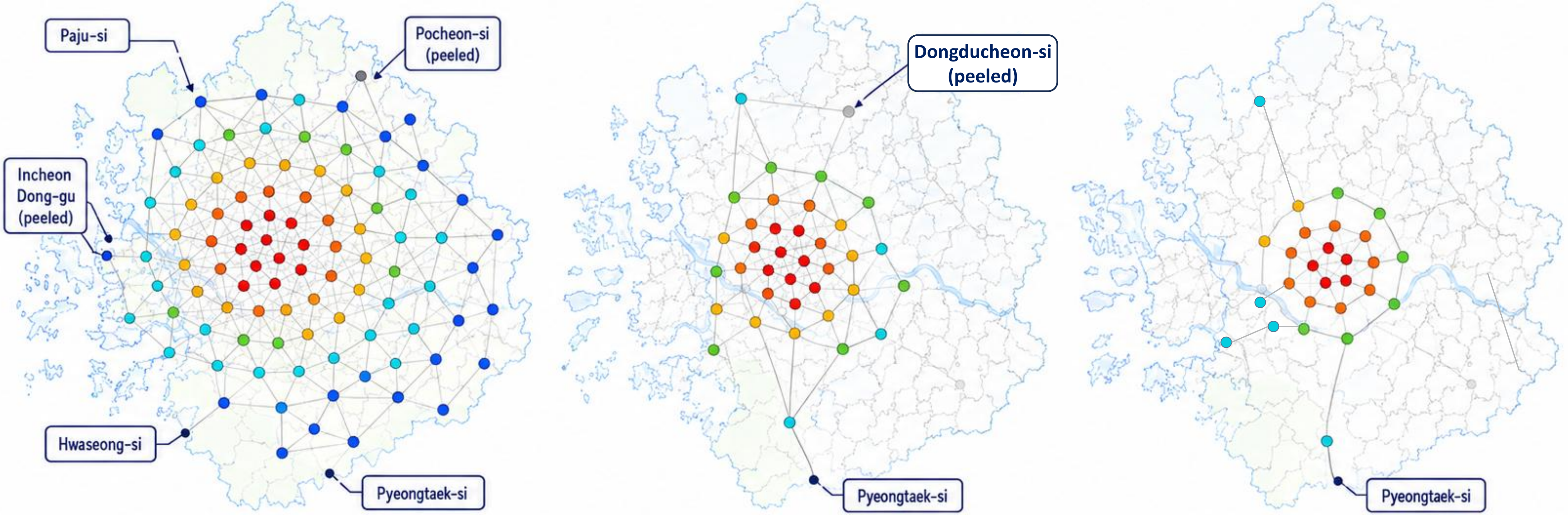
At near parity, the relation graph is sparse, but the remaining links carry a disproportionate share of demand.

Activation-shell erosion at θ = 0.9



Minimum activation α	Nodes retained	LCOC demand retention	FE retention
0.00	75	100.0%	100.0%
0.20	65	96.3%	79.6%
0.30	57	77.6%	62.6%
0.40	38	36.7%	29.4%
0.50	24	9.1%	12.3%
0.60	14	1.5%	4.2%

θ_c progressively thins the TCRON



core kept: 73 zones | largest core: 73 zones
links kept: 2,628

core kept: 55 zones | largest core: 55 zones
links kept: 502

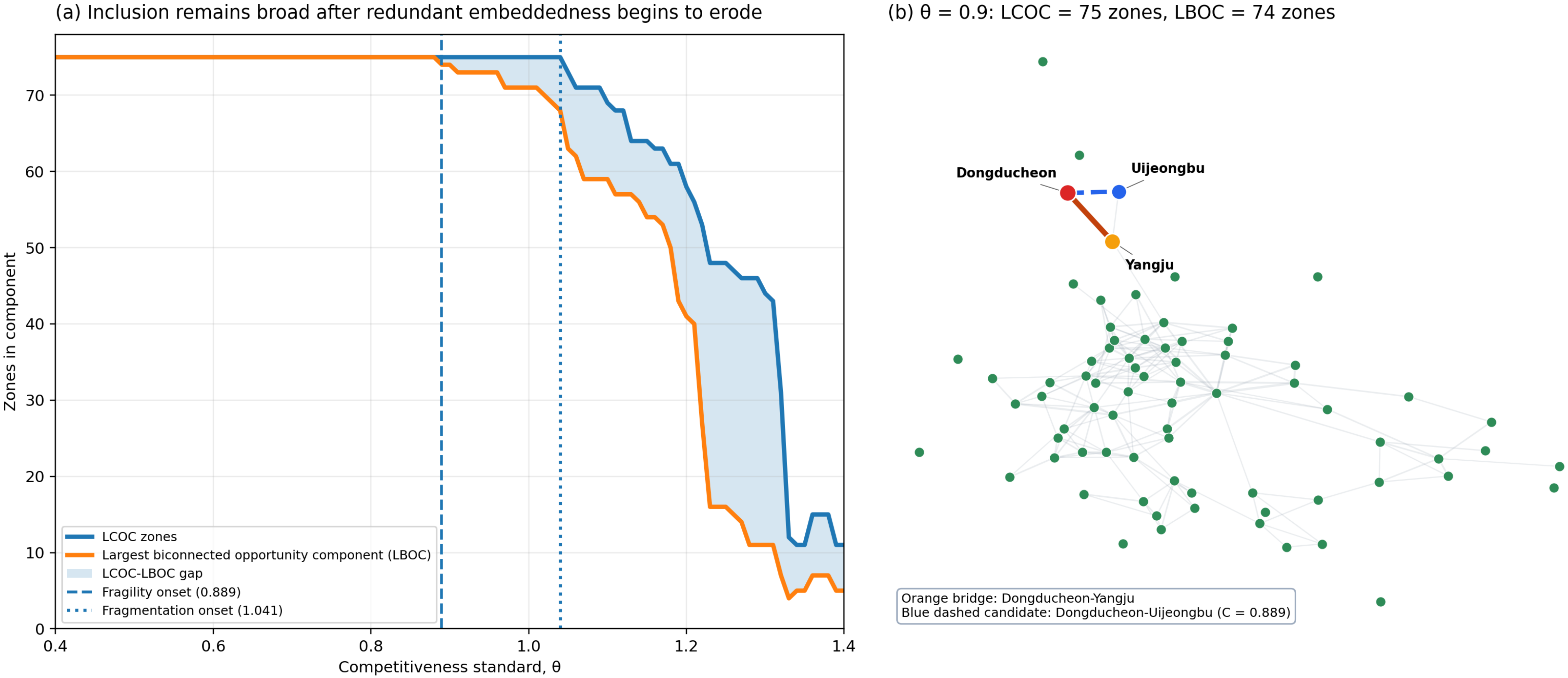
core kept: 51 zones | largest core: 47 zones
links kept: 91

Node activation
Low activation — High activation

Nodes and links in the pre-collapse core (kept at the selected cutoff)

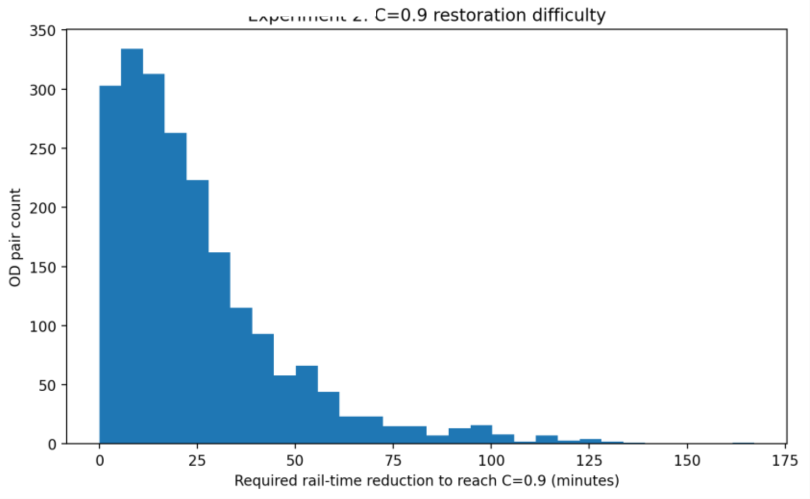
Peeled nodes and links (not in the core)

Fragility onset precedes component fragmentation



Many below-target $\theta_c (= 0.9)$ OD pairs can be restored with modest time savings

- Among below-threshold rail-trip weight, 36.8% requires at most 5 minutes of rail-time reduction, 62.1% at most 10 minutes, and 80.3% at most 15 minutes.



$$\Delta T_e^{0.9} = max\left(0,\ T_e^{rail} - \frac{T_e^{road}}{0.9}\right)$$

Required rail-time reduction	OD edges	Share of below-threshold edges	Share of below-threshold rail trips
$\leq$ 5 min	264	12.5%	36.8%
$\leq$ 10 min	563	26.6%	62.1%
$\leq$ 15 min	866	41.0%	80.3%
$\leq$ 20 min	1,132	53.5%	88.7%
$\leq$ 30 min	1,510	71.4%	96.7%
$\leq$ 60 min	1,965	93.0%	99.9%

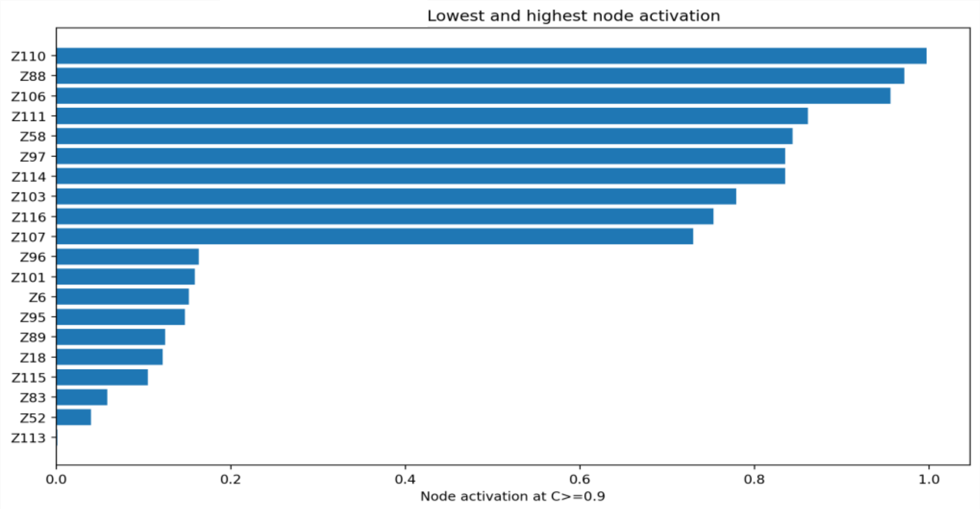
Impedance variables

- C is the criterion for determining rail competitiveness(filtering variable), while auxiliary time, transfers, and detour act as impedance variables explaining why rail fails to compete.

Variables	Why it is used here
Auxiliary ratio = $\frac{A}{T^{rail}}$	Diagnoses access, first/last-mile, and station-interface burden
Transfers = $\max(0, boardings - 1)$	Diagnoses transfer complexity and direct-service gaps
Detour excess = $\max(0, \frac{D_{rail}}{D_{road}} - 1)$	Diagnoses route indirectness and direct-service gaps

Node activation identifies where incident rail demand remains competitive

- Low activation does not mean no rail; it means much of a node's incident rail demand fails the high-competitiveness standard.



Group	Zone	Node activation	Main bottleneck	Planning implication
Lowest activation	Pocheon-si, Gyeonggi Province	0.1%	Access / auxiliary time	Improve feeder / first-mile
	Dong-gu, Incheon Metropolitan City	4.0%	Rail detour / indirectness	More direct service
	Uijeongbu-si, Gyeonggi-do	5.9%	Transfer burden	Reduce transfer burden
	Yeoncheon-gun, Gyeonggi-do	10.5%	Transfer burden	Reduce transfer burden
	Geumcheon-gu, Seoul Metropolitan City	12.2%	Transfer burden	Reduce transfer burden
Highest activation	Hwaseong-si, Gyeonggi-do	99.7%	Transfer burden	Reduce transfer burden
	Pyeongtaek-si, Gyeonggi-do	97.2%	Transfer burden	Reduce transfer burden
	Paju-si, Gyeonggi-do	95.6%	Transfer burden	Reduce transfer burden
	Gwangju-si, Gyeonggi-do	86.2%	Transfer burden	Reduce transfer burden
	Seo-gu, Incheon Metropolitan City	84.4%	Transfer burden	Reduce transfer burden

Below-target $\theta_c (= 0.9)$, failure is mostly a directness problem

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- For below-target OD pairs, rail detour and indirectness account for the largest share of failed rail-trip weight, followed by access burden and then transfer burden.
- Access improvements alone cannot restore most below-threshold OD pairs; directness and express-service interventions are the primary restoration priority.

Dominant bottleneck	OD edges	Below-threshold edges	Below-threshold rail trips	Mean C	Mean $\Delta T_e^{0.9}$	Aux. ratio	Transfers	Detour
Access / auxiliary time	761	36.0%	38.9%	0.750	23.2	0.390	1.47	1.029
Transfer burden	191	9.0%	2.7%	0.729	23.9	0.153	2.76	0.947
Rail detour / indirectness	1,162	55.0%	58.4%	0.701	25.9	0.262	1.76	1.368
Already C ≥ 0.9	661	n.a.	n.a.	1.086	0.0	0.393	1.24	1.100

Planning implications: differentiated intervention portfolios

Protect demand backbone

Target

High-strength relations whose loss has system-wide consequence

Policy package

Capacity, reliability, disruption management, service protection

Repair tethered inclusion

Target

Zones included through one articulation structure

Policy package

Candidate second competitive tie; corridor review

Strengthen weak cross-regional relations

Target

Cross-regional relations with low competitive support

Policy package

Through-running, express patterns, transfer reorganisation, new links

Concluding remarks

- This study employed a modified k-core decomposition to examine how time-competitive the rail O/D network of the Seoul metropolitan area is relative to cars.
- The analysis shows that rail's detour characteristics relative to roads are the primary bottleneck, and that many O/D pairs can satisfy policy requirements with only modest improvements in rail travel time.

Time-competitive rail opportunity network → **Modified K-core decomposition** → **Network robustness test** → **Restoration prioritization**

Thank you!

Q & A

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Latent-class mode choice as behavioral impedance

- The latent-class model serves as a behavioral impedance and weighting device rather than a replacement for the C threshold.

$$V_{ng} = \beta_{g0} + \beta_{gC} C_{ij} + \beta_{gA} \mathrm{AuxRatio}_{ij} + \beta_{gH} \mathrm{Transfers}_{ij} + \beta_{gR} \mathrm{DetourExcess}_{ij} + \beta_{gX} X_n$$

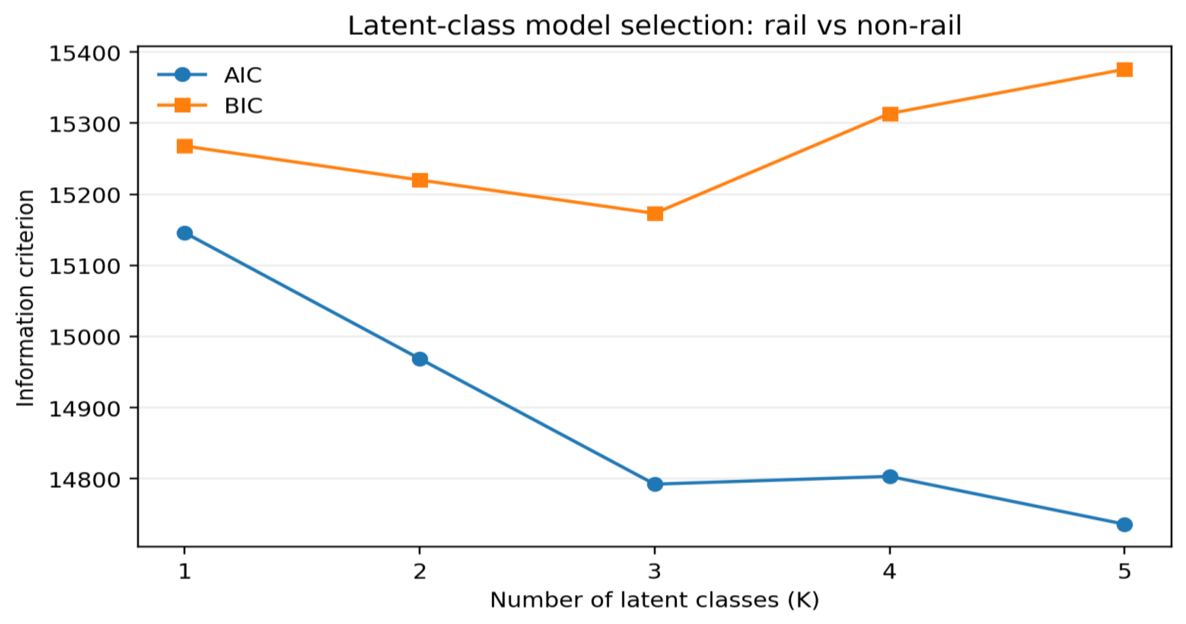
$$P(\, y_n = \mathrm{rail} \mid g\,) = \Lambda\left(V_{ng}\right)$$

$$P(y_n = \mathrm{rail}) = \sum_{g} \pi_{ng} P(\, y_n = \mathrm{rail} \mid g\,)$$

$$W_e^{\mathrm{LCM}} = \sum_{n \in e} \hat{P}(y_n = \mathrm{rail})$$

Latent-class model results

Class	Interpretive label	Posterior share	Observed rail share	Predicted rail prob.	Mean C	Aux. ratio	Transfers	Detour excess	Car available
1	Rail-reluctant / non-rail-oriented	36.3%	15.3%	15.9%	1.014	0.568	0.92	0.378	81.5%
2	Intermediate multimodal users	37.0%	24.5%	26.5%	1.012	0.566	0.93	0.378	81.0%
3	Rail-inclined / rail-receptive	26.7%	46.6%	57.6%	1.008	0.556	0.94	0.375	82.6%



High-competitiveness rail is sparse but demand-concentrated

C threshold	Edges	Edge retention	Rail-trip retention	Purpose-demand retention	LCM rail potential	Largest component nodes	Functional efficiency
0.5	2,741	98.8%	99.7%	99.4%	99.6%	75	0.994
0.7	1,916	69.0%	86.6%	85.1%	84.7%	75	0.845
0.9	661	23.8%	40.8%	52.0%	40.7%	75	0.607
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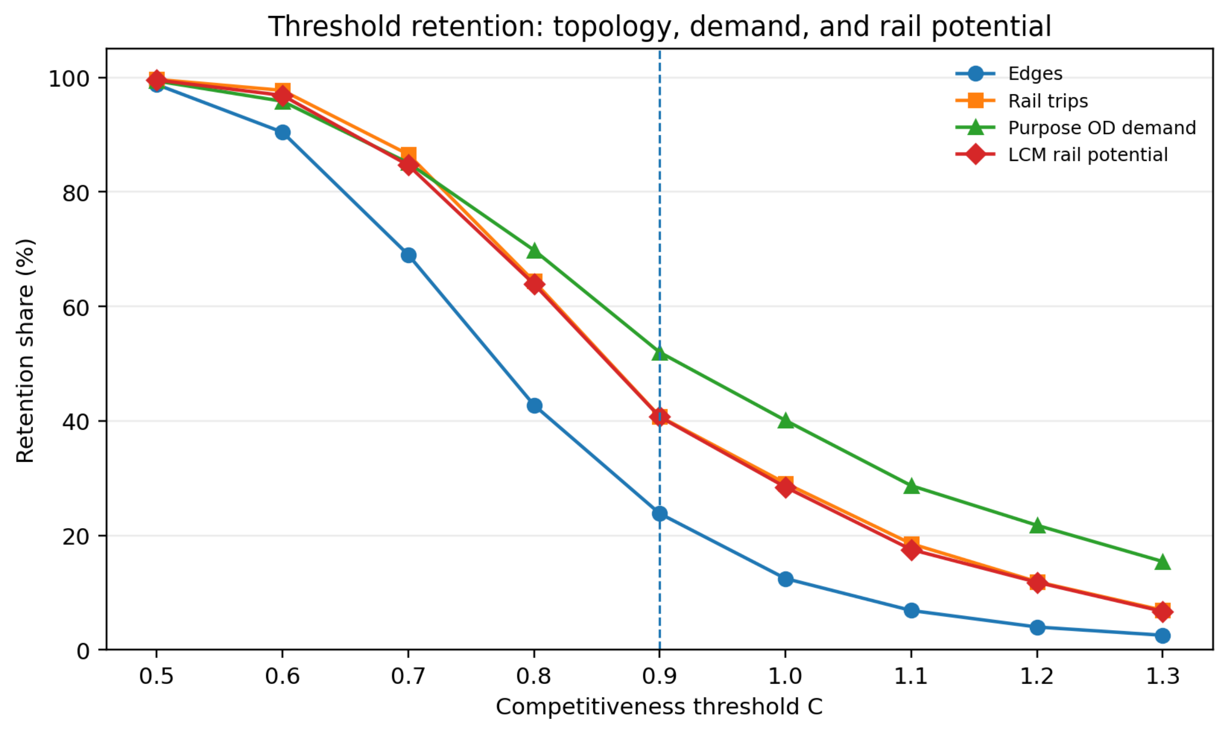
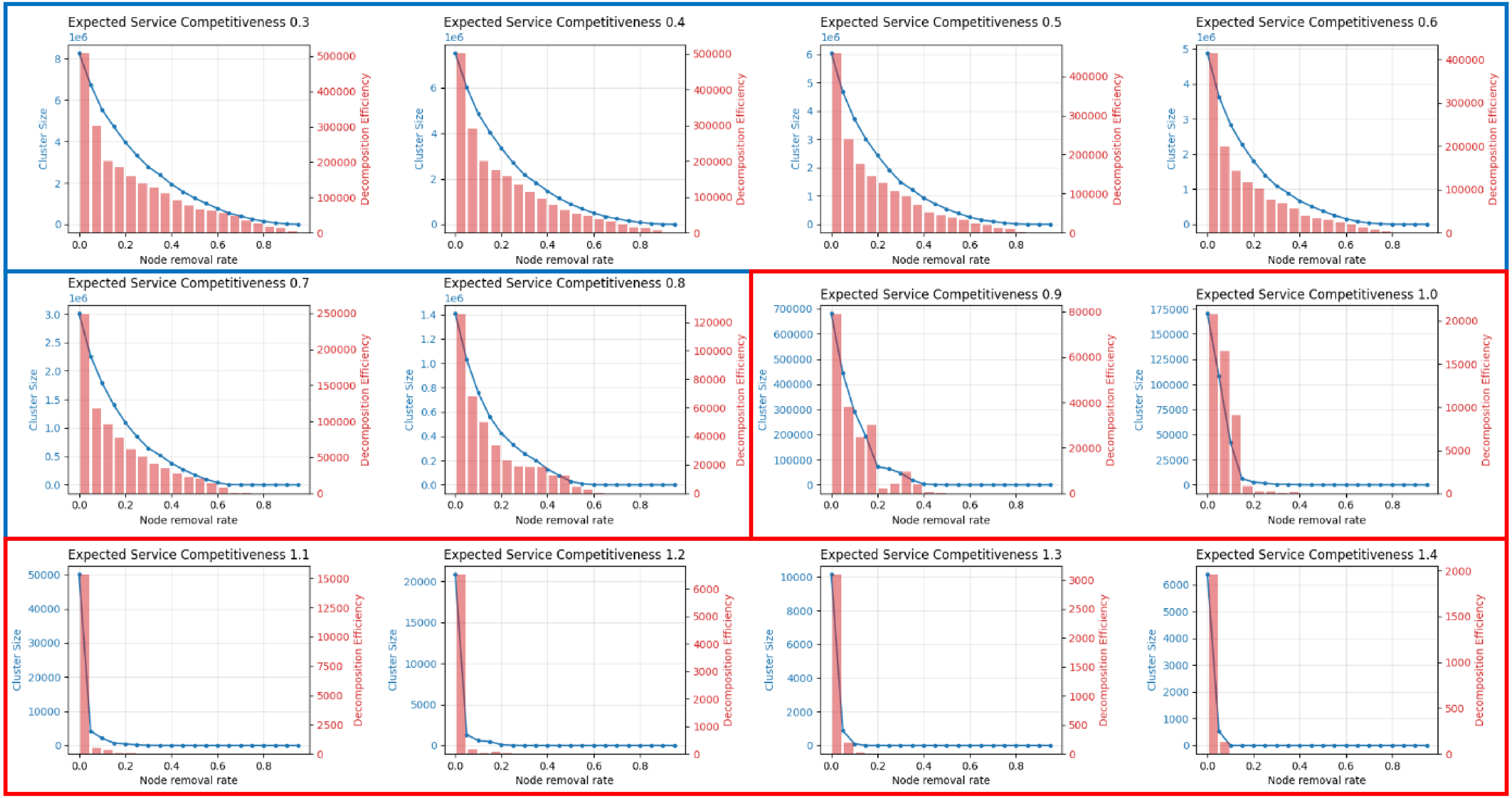
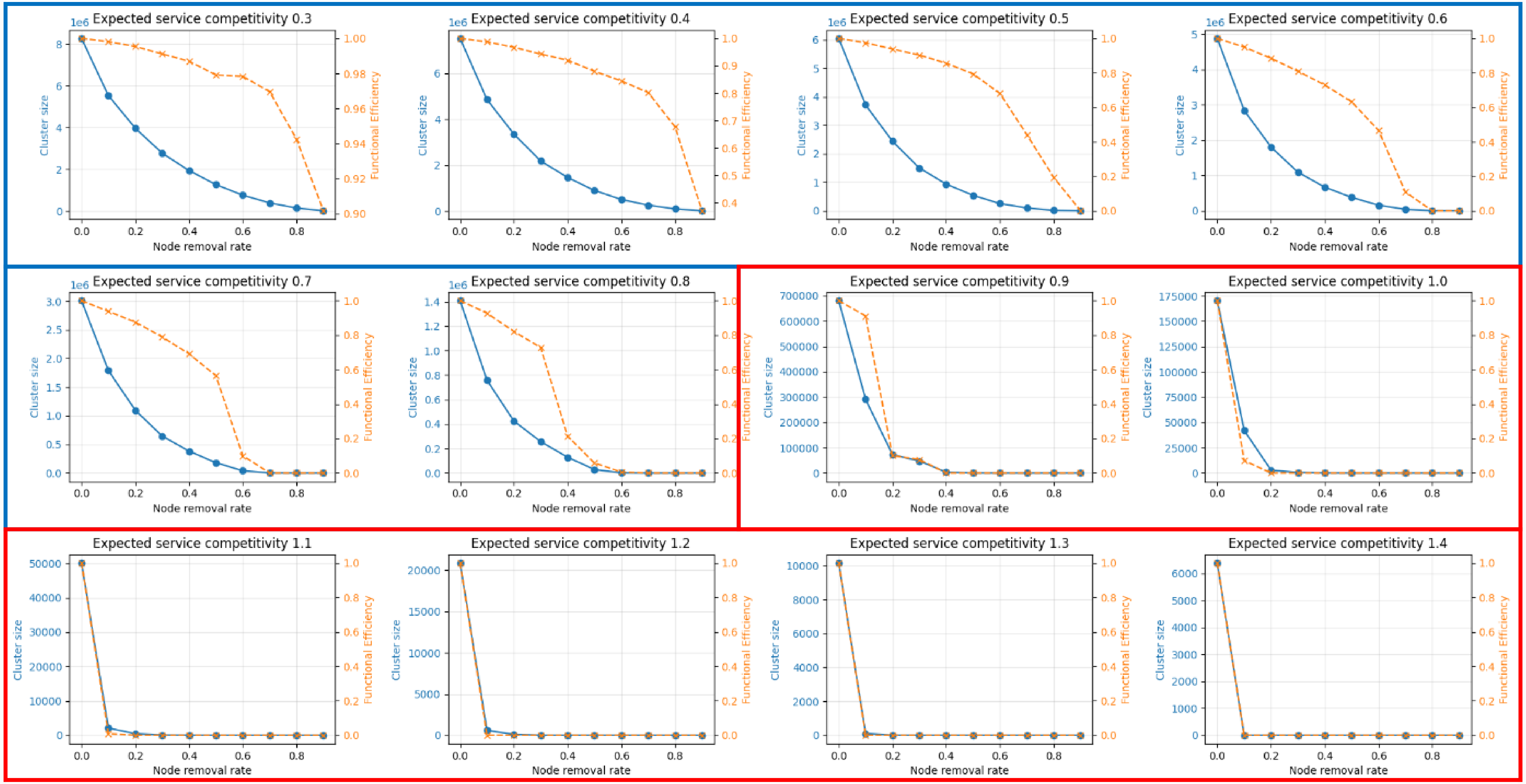


Robustness under increasing competitiveness thresholds

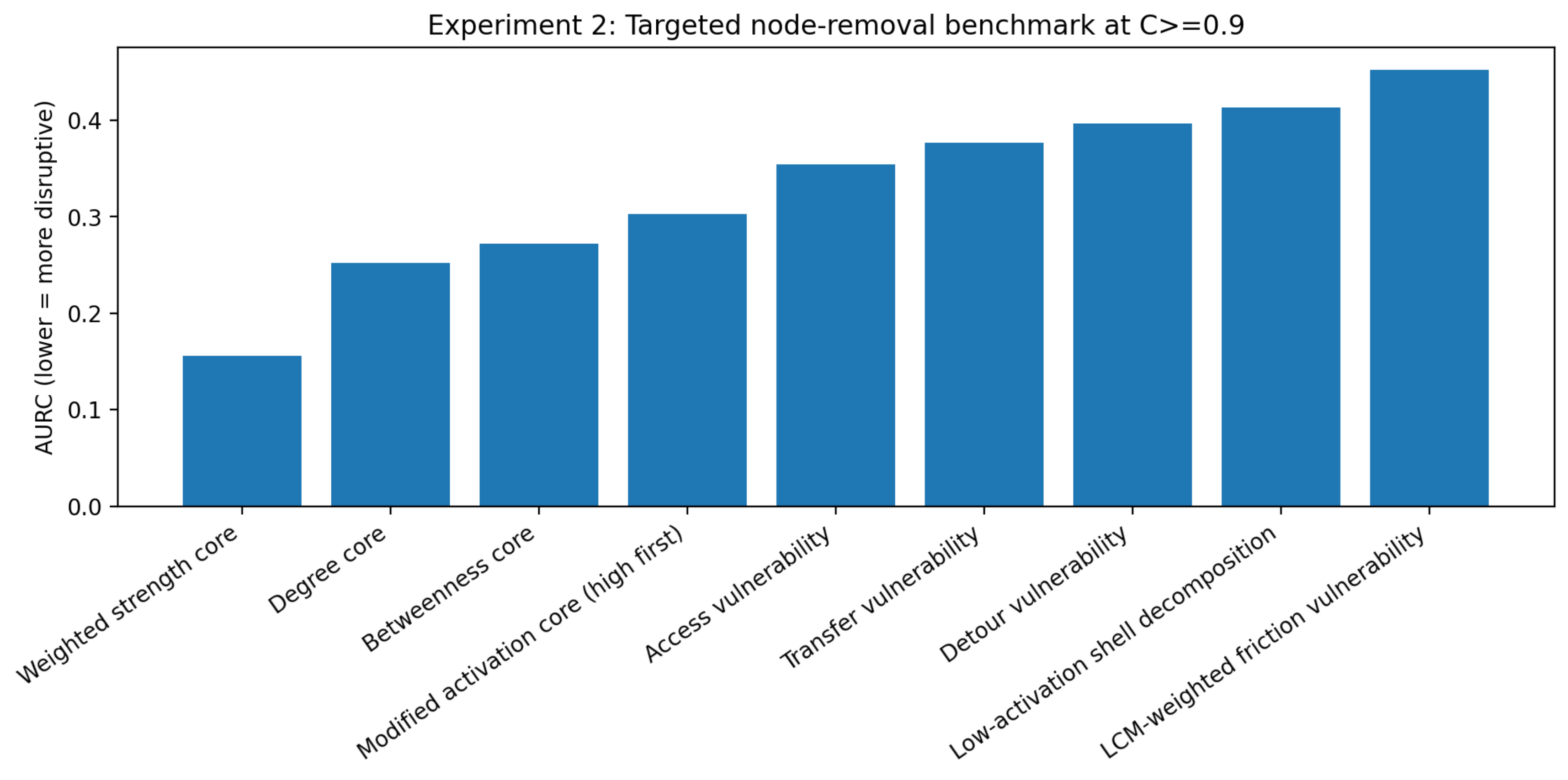


Functional efficiency under increasing competitiveness thresholds



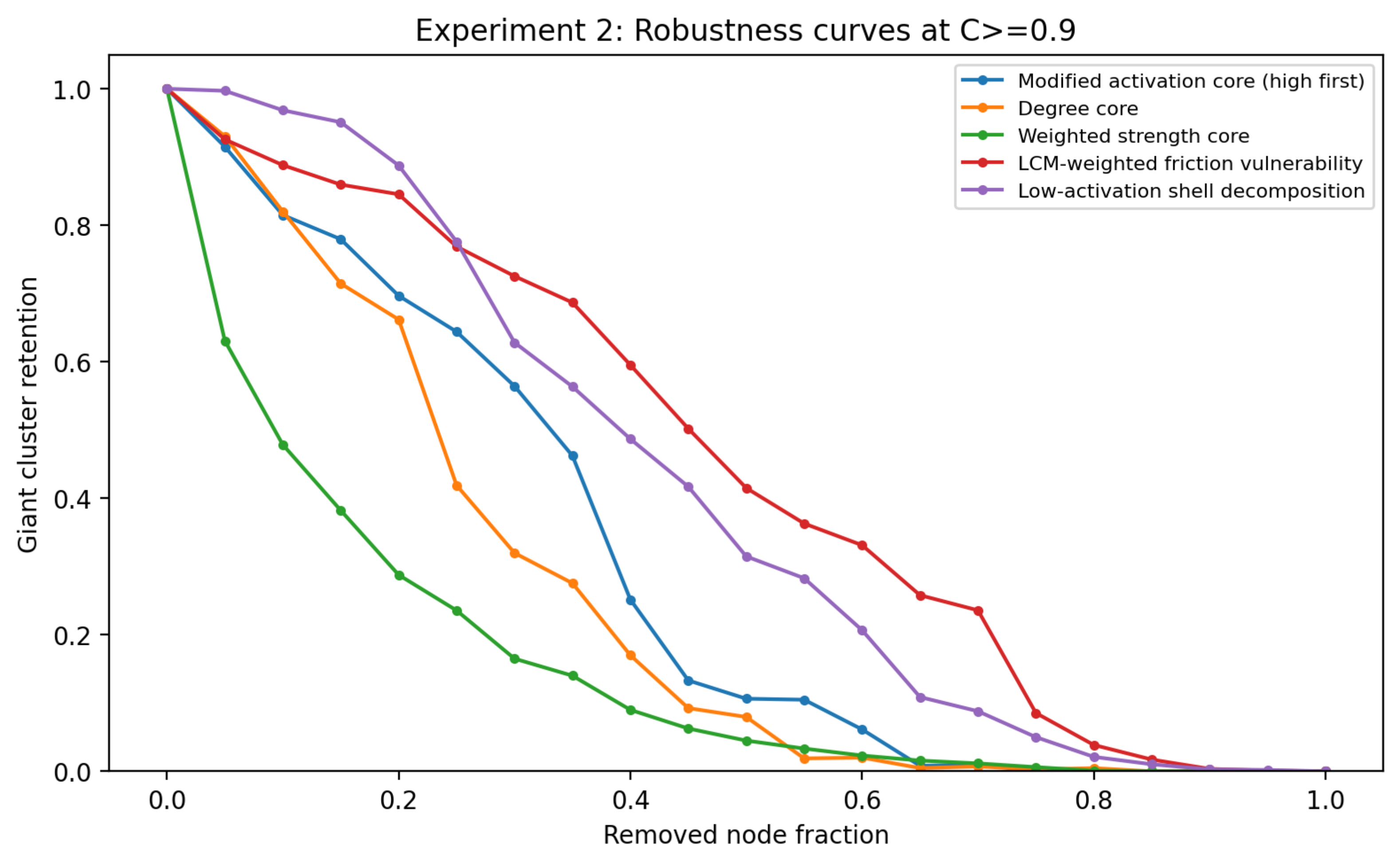
Benchmarking prevents overclaiming the modified k-core

- Weighted-strength removal is most disruptive, while activation-based decomposition reveals demand-functional layers rather than simply identifying the fastest attack.



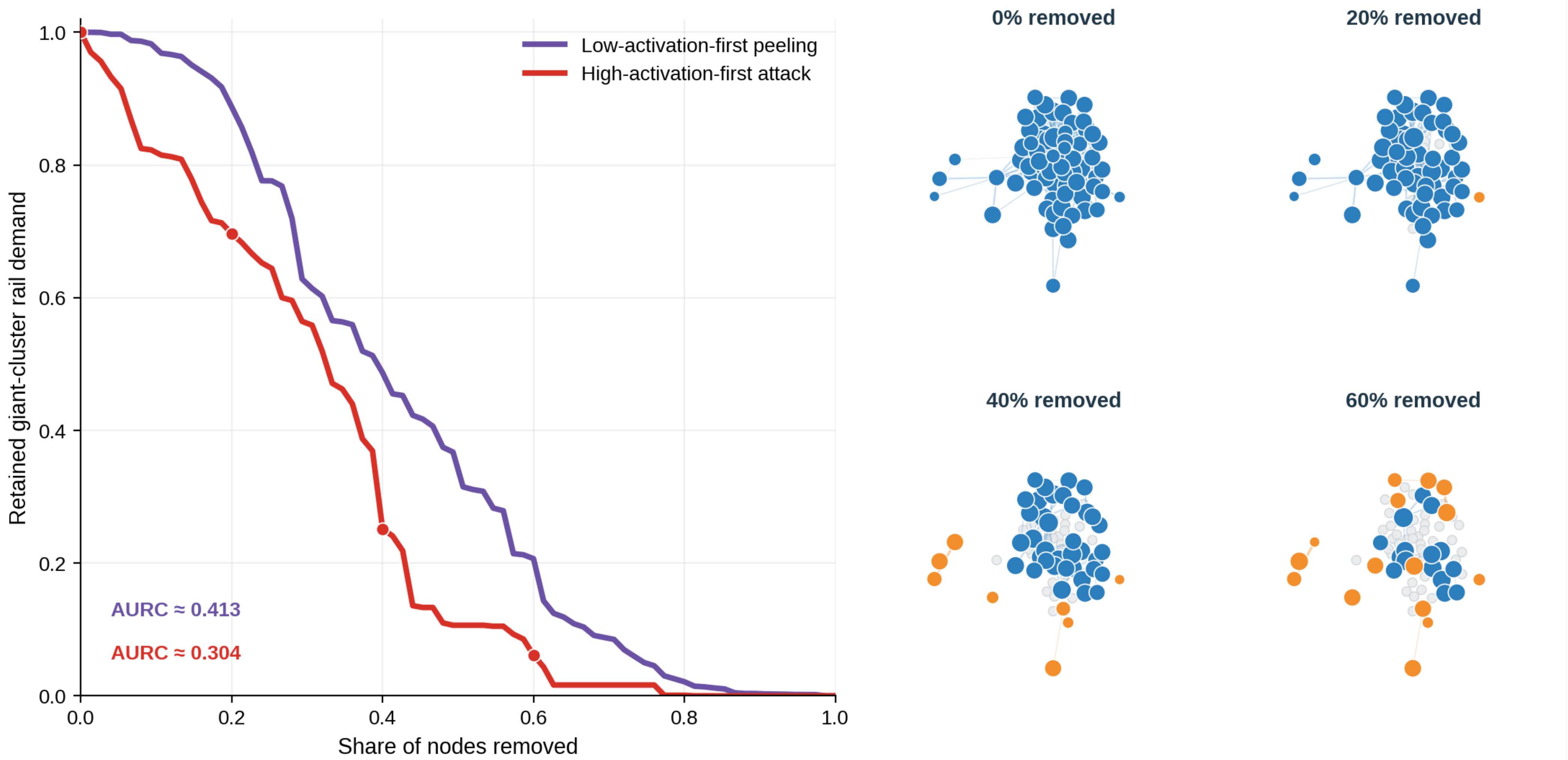
Robustness curves separate structural criticality from service-friction vulnerability

- Central demand hubs and high-friction zones are not the same objects; opportunity-network robustness requires both structural and service-friction readings.



Network snapshots explain the AURC robustness curves

Low-first peeling reveals functional layers, whereas high-first removal tests structural criticality.



The images expose what the curve aggregates: a connected graph can lose most demand-functional cohesion before all nodes disappear.

Author calculations from uploaded KTDB OD skims; visualization adapted from Alvarez-Hamelin et al. (2005).